simon-5501-01-slides

2026-08-25

Topics to be covered

- What you will learn
  - About this class
  - R and RStudio
  - History of R
  - Scales of measurement
  - Tests of hypothesis
  - Confidence intervals
  - A simple R program
  - Your homework

Welcome to the class

- Three sections
  - 0001, Synchronous Zoom meetings on Tuesdays
  - 0002, Asynchronous
  - 0003, International students
- Introductions
  - Ricardo Moniz
  - Suman Sahil

Speaker notes

There are three sections in the class.

Section 0001 is a synchronous online class with lectures delivered via Zoom on Tuesdays from 1:00pm to 3:45pm. Students in this section are required to attend that class. If you are in this section, you must show up on Tuesday afternoons. Attendance will be taken. Two or more unexcused absences will require a discussion with the instructor and remedial actions may need to be taken.

Section 0002 is an asynchronous online class where you watch videos of the zoom session. Students in this section have the option of attending the Tuesday 1pm Zoom sessions, if their work schedule permits.

Section 0003 is a synchronous online class like section 0001 with the additional requirement of weekly sign-ins. The weekly sign-in meets an important requirement of some international students. Other than the sign-in, the class is identical to the synchronous online class.

All three sections are combined in Canvas to a single site. This makes it easier for me to make announcements, do grading, etc. Just because the Canvas site says “2024FS-MEDB-5501-0001” does not mean that you are in the wrong Canvas site and it does not mean that you are registered for the wrong class. We’re all one big family from the perspective of Canvas. Only the attendance requirements differ.

Requirements of all students

- Attend Tuesday Zoom or watch video of Tuesday Zoom
- Read book chapter
- Optional review session on Zoom on Fridays
- Complete all assignments by Monday at 11:59pm

Speaker notes

I'll record the Tuesday 1pm meeting and make it available on Canvas by noon on Wednesday for the asynchronous students.

Everyone should also read the appropriate Chapter from the book.

Everyone is invited to an optional review session on Fridays from 10:30am to 11:30am. You can ask questions, or just sit back and listen to questions that others may have. If you have work commitments on Friday mornings, I am still happy to meet with you at any other reasonable time.

The homework assignments, quizzes, and discussion boards are due by 11:59pm on Mondays.

Attendance

- Synchronous students
  - MUST attend Tuesday Zoom sessions.
  - can also review the recordings
- Asynchronous students
  - watch the Tuesday recordings
  - or attend some of the Tuesday Zoom sessions
  - or both
- Failure to attend is a problem

Speaker notes

Attendance at the Tuesday Zoom sessions is a requirement for all synchronous students. You can also watch part or all of the recordings of the Tuesday Zoom sessions if you like.

Asynchronous students have an option that synchronous students do not.

For synchronous students to attend the Tuesday Zoom sessions is a problem. For asynchronous students, failure to either attend or watch the videos is a problem. If you have work commitments, health issues, or personal issues that prevent you from meeting the attendance requirements, you must contact us, preferably beforehand.

Encouraging class participation

- Synchround students have more opportunities
  - Raised hand icon
  - Chat window
  - Just unmute yourself

Speaker notes

If I had a fault as a teacher, it would be that I talk too much. Not that the stuff I talk about is bad, but it’s difficult to get good student interactions.

In a zoom session, it’s hard. But certainly those of you who are watching asynchronously to my videos, it’s even more difficult. We certainly encourage you to use the discussion boards and there’s some interaction there.

But those of you who are live, you have a special opportunity. Anytime something’s confusing, raise your hand. There’s a little icon where you can raise your hand.

Or post something in the chat box. I have the chat window open.

Or just unmute yourself and start talking.

Uh, I will tell you right now that, uh, a lot of students are bashful and they think, uh, no one cares about my question. And that’s almost certainly false, because when someone asks a question, it’s usually 2 or 3 other people who have the exact same question, but they haven’t figured out how to articulate it, or they’re even shyer than you are.

So please, please, if you are here, live, um, ask questions.

Those of you who are listening asynchronously, you can’t ask questions during the video, but you can show up at the help sessions on Friday morning.

You can post questions on the discussion board. We always ask for feedback on the discussion board

Or just drop an email if you need to ask a question. I am happy and Suman is happy to meet with you at any time to help you with this class.

My favorite part of the job is interacting with students and you don’t get to do that in an asynchronous online class very much.

Even with, uh, those people listening live with zoom–Zoom’s different.

When you are talking on Zoom, you can’t look around the room, you can’t, walk around, you can’t make grand gestures. That doesn’t work. What’s worse is that you miss a lot of the capabilities of reading the room–noticing when students are confused.

So don’t be afraid to ask for help. I hate to say this. The students that I like the best are the ones that struggle the most. Because I get to know them. And I get to interact with them. And it’s fun. Helping people through something difficult is something I really enjoy.

The students who are just breezing the class never have any problems, I’m glad for them. But they are total strangers to me.

Okay. So do not be bashful about asking for help.

Don’t be scared about R

- Almost all computer assignments will use the R programming language
  - Early examples will be “sppon-fed”
  - R is easier than you might think.
- Don’t expect miracles by the end of this class
  - You will not be an expert with R
  - You will be more comfortable with R

Speaker notes

I’ve always get comments about being a little scared about having to run this R program. I’ve never programed an hour before.

Suman and I will make it easy. The first time we tried this I was a little bit worried, but it went smoothly. It is not that hard to learn how to program in R.

The first few assignments go very, very slowly and there’s very little that you have to do.

You just make some minor modifications to an existing program.

Gradually, you will start to make bigger modifications.

At the end of the class, you will not be an expert in R, but you will be more comfortable with it and be able to learn more on your own.

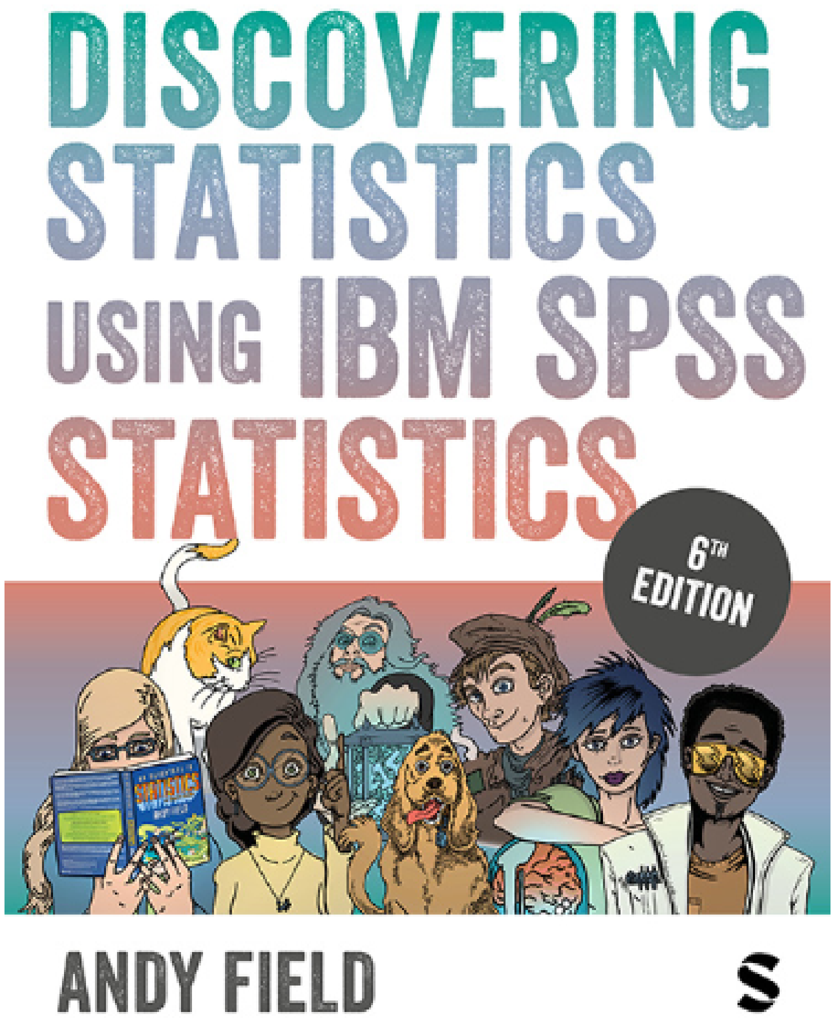
What about MEDB 5507, Introduction to R?

- That class places emphasis on data management
- This class places emphasis on data analysis

Speaker notes

I teach another class, MEDB 5507, Introduction to R. I want to emphasize that there is not too much overlap in these two classes. MEDB 6607 places the emphasis on programming and data management. MEDB 5501 places the emphasis on data analysis and data interpretation.

Your book


DISCOVERING STATISTICS USING IBM SPSS STATISTICS
6TH EDITION
ANDY FIELD

Speaker notes

There's a book that we have, written by Andy Field, a real nice guy. The book is written a little bit flippantly and that turned some people off.

I've been told by people who know more than I do that when you write really outrageously and come up with all these bizarre scenarios that it makes things more memorable. Andy Field does outrageous better than anyone.

It's a bit unfortunate because this book shows all the examples in SPSS. There is an earlier edition that uses R. You are welcome to use that book instead. It is organized a bit differently and misses a few topics, but the book is still good enough.

Either edition is very good. I can't take credit for this. The book was the one recommended by the person who used to teach this class, Monica Gaddis.

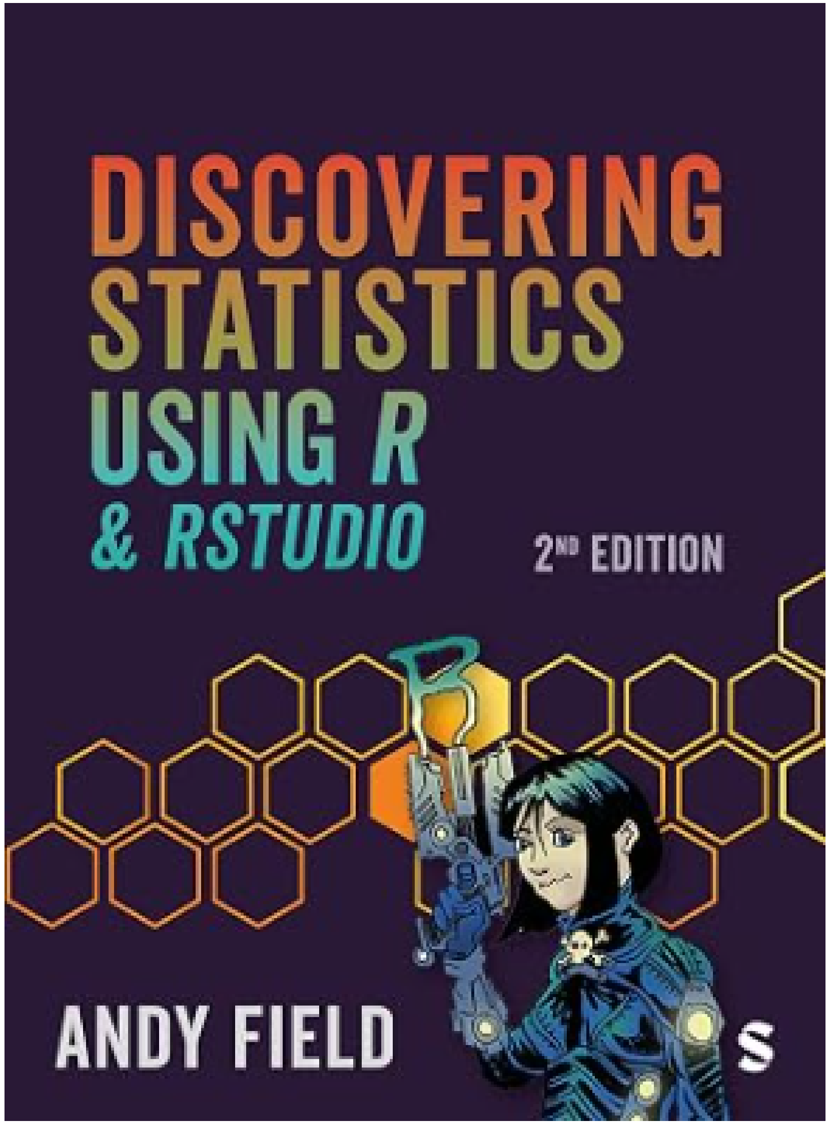
She liked it because it is very comprehensive. It doesn't cover everything, but it comes a lot closer than just about any other book I have seen. Monica described it as the only statistics book you'll ever need.

And she's not too far from the truth on that. If you are doing something statistical, you're likely to find it in this book.

Now there will be times when I disagree with how Andy Field presents things. He loves effect sizes for example and I hate them. That's okay. His voice is often more mainstream than mine.

I am not bashful about expressing my opinions, but I will let you know when other experts, especially Andy Field, have a different opinion.

Stop the presses!!!



Speaker notes

Stop the presses! When I was getting an image file of the book cover for Andy Field’s book, I noticed that there is a brand new version that just came out in July of this year. If you have not yet bought the SPSS version of Andy Field’s book, buy this one instead. I don’t have a copy yet myself, but it looks like it will be a winner.

Assignments

Speaker notes

Assignments will include homework submissions, quizzes, and discussion boards. Everything is due on Mondays at 11:59pm. Short term exensions are easy.

This class is in transition

- Taught for many years by Dr. Monica Gaddis
- My third time teaching this class
- Major changes
  - Software agnosticism
    - In class switch from SPSS to R
  - Suman Sahil will co-teach and cover programming in R
  - Discussion boards ask for feedback
  - Proposed exam questions

Speaker notes

I’m excited at the opportunity to teach this class, but I am also a bit scared. The topics in this class are relatively easy, but teaching them well is not so easy.

This class has been taught for many years by Dr. Monica Gaddis. She is an excellent teacher, but she retired this summer. I have a lot of respect for her and have kept a lot of her material. There are some important differences, however. I am not a wall flower, and I have a few ideas that might supplement an already good class.

I believe in software agnosticism. You should use whatever software you thing is best for you. Use the software that you expect to use in the real world after you graduate. It’s more work for me (and for Suman Sahil, who has to grade the various programs). Nevertheless, I want to make that effort because I should not presume to know what program is best for you and your budding career.

All the classroom examples will use R. If you use another program, expect to spend some time reading the documentation and figuring out how to convert the R code to code for your system. You can ask for assistance by email, during the help session if all the other questions are already answered, or by special appointment. I take great pride in knowing how to do things in a broad range of statistical software and should be able to help with most issues.

If you’ve never used any statistical software before, I would certainly encourage you to use R.

A second change is that I will include a brief feedback assignment after every module. I want to feedback on what you thought was most important, what you found most confusing, and what you might have liked to learn that wasn’t in the lecture.

Don’t spend a lot of time on the feedback. You are welcome to say something brief like nothing was confusing. If everything was confusing, that’s fine also, but please feel free to show up at the Friday review session or set up a special appointment if this is the case.

You are also welcome to say something like “I agree with” and mention the name of a previous student.

I might make some other small changes over time.

There’s a risk in making changes. It may create some inconsistencies between what I am asking you to do and what Dr. Gaddis covered in her videos. Use a bit of common sense, but if you are not quite sure about anything, please ask.

Why R

- Faculty consensus
  - Prepares you better for capstone/thesis work
  - More likely to be used in your future job
  - Integrates well with team based tools
- R is not as difficult as claimed
  - Work from existing templates
- Python, SAS, Stata would have also been good choices
  - You are welcome to use these if you like

Speaker notes

Last year, I taught the class using SPSS and it was an okay choice. But some of the faculty raised concerns about leaving students ill prepared if they needed to do data analysis for a capstone project or thesis. SPSS is okay for introductory statistics, but falls short for the sort of work needed in a thesis.

While some jobs out in the real world will use SPSS, far more will use R.

Most important, in my opinion, is that R integrates well with team-based tools like version control, cloud-based computing, distributed databases, and other methdologies that are being used more and more with programming teams.

You may have heard that R is a difficult language. R is not as difficult as some people may claim. It is being used successfully at a large number of universities in their Statistics and Data Science program. It is even used extensively in undergraduate classes.

Having said that, I have to admit that R is still not as easy to use as SPSS. I will try to lighten that burden by providing templates that will guide your coding throughout all of the assignments.

Now there are several good alternatives to R. Each has its advantages but also some disadvantages. It is not worth quibbling here. I've chaired a couple of panel discussion sessions at national Statistics conferences on this topic. My recommendation is to use what your boss uses.

If you want to use something other than R, go for it! You'll have to do a bit of work on your own, but I will provide support as needed. I take great pride in being able to work with a broad range of statistical software systems.

Student learning objectives, 1 of 3

- SLO1
  - The graduate will be able to use statistics to analyze and interpret data. They will understand the fundamentals of the field in the context of recognizing the effective use of data or information for the specific discipline(s). They will select and apply appropiate statistical procedures to the information. They will be able to analyze and accurately interpret of statistical result.

Speaker notes

There are five student learning objectives that our department has committed to. We need to demonstrate to the university that anyone who graduates from our department has a particular set of capabilities. This class is one of several that provides you with these capabilities.

The first student learning objective asks that you are capable of using statistics to analyze and interpret data. This is not the only class where you will develop this capability, of course. The key elements are selecting the statistical procedures, running those procedures, and writing about the statistical procedures, and the ability to run those procedures.

Note: the Student Learning Outcomes spreadsheet has three entries: Select, Write, and Run.

Student learning objectives, 2 of 3

- SLO2
  - The graduate will be able to design a testable research question or hypothesis. They will have adequate background knowledge about biological, biomedical, or population health contexts and problems including common research problems in order to generate a research question or hypothesis. They will be able to relate problems within and across levels of areas of the spectrum to bridge disciplines.

Speaker notes

The second student learning objective is the ability to design a testable research question or hypothesis.

Note: the Student Learning Outcomes spreadsheet has one entry: Hypothesis.

Student learning objectives, 3 of 3

- SLO5
  - The graduate will be able to communicate scientific outcomes. This includes the ability to convey scientific methods and statistical findings, effectively field questions in an oral presentation format as well as in the preparation of thesis or capstone manuscripts.

Speaker notes

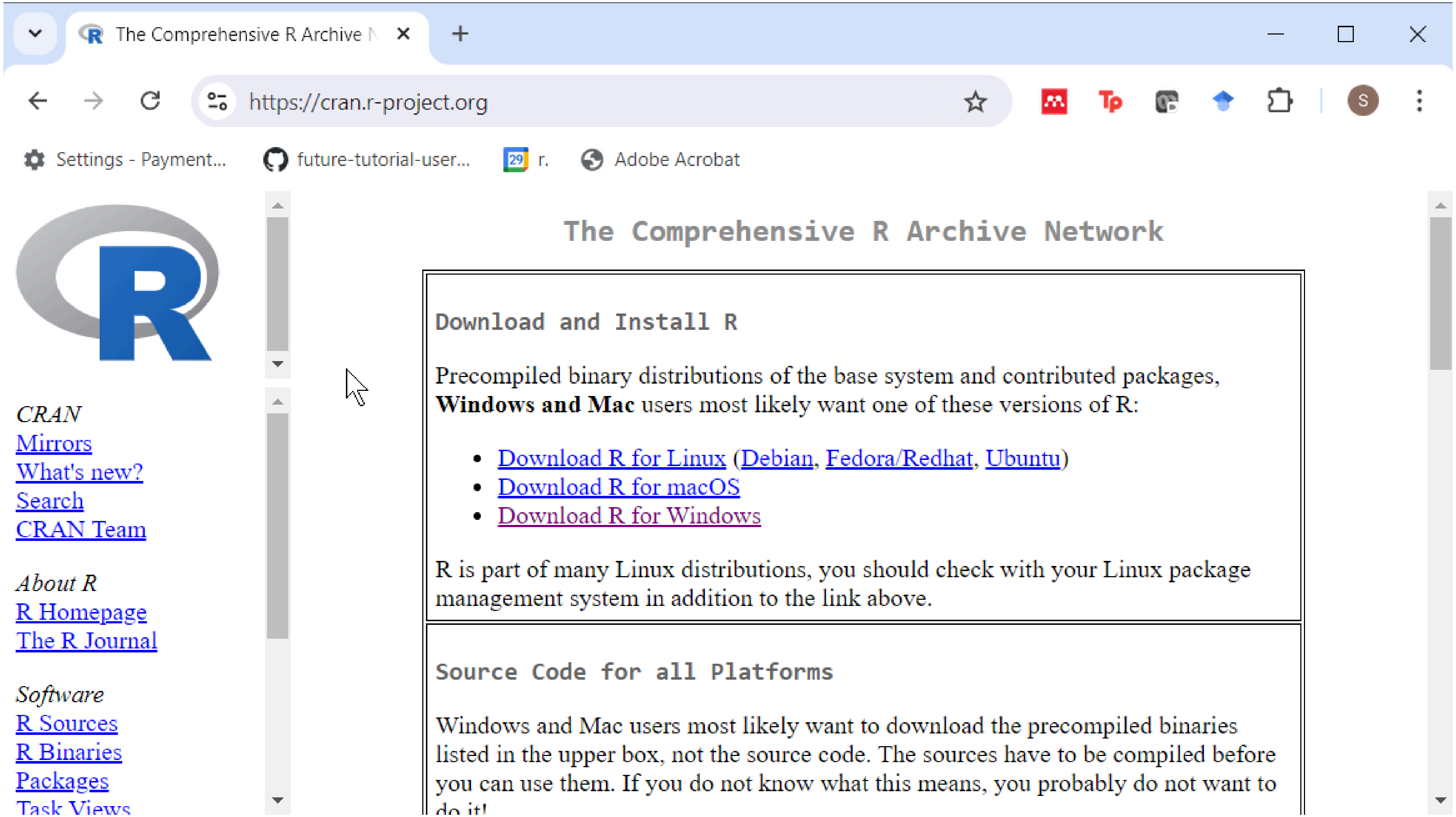
The third and fourth student learning objectives are developed in other courses. The fifth (and last) learning objective is the ability to convey statistical findings and answer question in an oral presentation. This includes the ability to present your findings in a visual format.

Note: the Student Learning Outcomes spreadsheet has two separate entries: Results Visual and Results Write-up

Break #1

- What you have learned
  - About this class
- What's coming next
  - R and RStudio

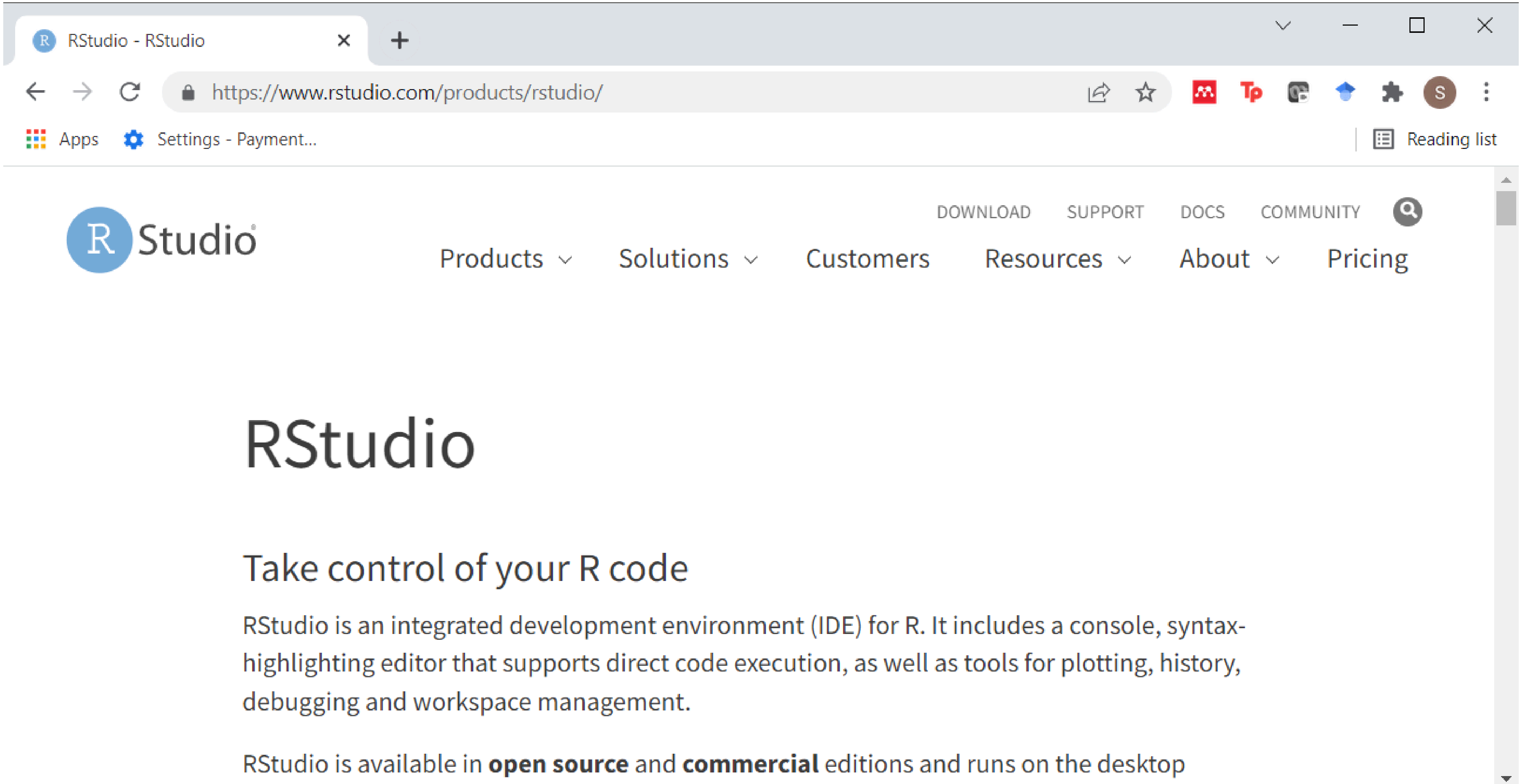
R


The Comprehensive R Archive N
https://cran.r-project.org
Settings - Payment...
future-tutorial-user...
r.
Adobe Acrobat
The Comprehensive R Archive Network
CRAN
Mirrors
What's new?
Search
CRAN Team
About R
R Homepage
The R Journal
Software
R Sources
R Binaries
Packages
Task Views
Download and Install R
Precompiled binary distributions of the base system and contributed packages, Windows and Mac users most likely want one of these versions of R:
Download R for Linux (Debian, Fedora/Redhat, Ubuntu)
Download R for macOS
Download R for Windows
R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.
Source Code for all Platforms
Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to

Speaker notes

This course will use R for all classroom examples and programming assignments. You can download R from CRAN, the Comprehensive R Archive Network.

RStudio


RStudio - RStudio
https://www.rstudio.com/products/rstudio/
Apps
Settings - Payment...
Reading list
RStudio
DOWNLOAD
SUPPORT
DOCS
COMMUNITY
Products
Solutions
Customers
Resources
About
Pricing
RStudio
Take control of your R code
RStudio is an integrated development environment (IDE) for R. It includes a console, syntax-highlighting editor that supports direct code execution, as well as tools for plotting, history, debugging and workspace management.
RStudio is available in open source and commercial editions and runs on the desktop

Speaker notes

RStudio is an integrated development environment. This provides a window where you edit your code, a window to show your files, a window to display your text output, a window to display your graphs, and other helpful windows.

Other statistical software

- JMP, Python, SAS, SPSS, Stata
  - Use only if you are confident in your abilities
  - Avoid Microsoft Excel

Speaker notes

There are lots of other choices. I am happy to help you get started with any of these programs. I like working with software. I should warn you that the level of support that I can provide does vary. I know less about Python and Stata, for example, than I do about SPSS and SAS. But I’m pretty good with anything.

If you do decide to try one of these programs, make sure that you are confident in your own abilities and that you can figure out some stuff on your own. If you are new or relatively inexperienced, SPSS might be a better choice.

Live demo

Break #2

- What you have learned
  - R and RStudio
- What’s coming next
  - History of R

History of R


History of R – Pmean
Ask Gemini
Not secure
pmean.com/posts/history-of-r/
Pmean
History of R
*BLOG POST
YEAR 2014
R SOFTWARE
AUTHOR
Steve Simon
PUBLISHED
May 30, 2014
R sprouted from S
NEW
The S Language
A PROGRAMMING ENVIRONMENT FOR DATA ANALYSIS AND GRAPHICS
Richard A. Becker
John M. Chambers
Allan R. Wilks

Speaker notes

I want to talk a little bit about the history of R. I think it’s probably, a fault of old people. We’re so obsessed about history because we live through so much of it. But it is important to understand where R came from or where SAS came from or SPSS or Python. If you understand those things, then you’ll understand why they made choices they did and how they fit in today.

Break #3

- What you have learned
  - History of R
- What's coming next
  - Scales of measurement

Scales of measurement, 1

- Dichotomy
  - Continuous
  - Categorical

Speaker notes

There is a basic dichotomy of data types: categorical and continuous. Now I have said many many times before that I hate dichotomies. Nevertheless, they are useful.

Scales of measurement, 2

- Stevens scales of measurement (controversial!)
  - Nominal
  - Ordinal
  - Interval
  - Ratio
- Addition/subtraction not allowed for ordinal data
  - Mean of ordinal data is meaningless

Speaker notes

A psychologist, Stanley Smith Stevens divided the entire universe of data into four categories: nominal, ordinal, interval, ratio. I won't review the definitions for all of these, but ordinal data is categorical data where there is a natural ordering of categories. An important limitation to ordinal data, but where the spacing between successive units is not consistent.

The belief among many (but not all) researchers, is that certain statistics are inappropriate for certain measurement scales. In particular, these researchers are very fussy about ordinal data.

An example of ordinal data.

- “Do you agree or disagree with the following statements”
  - “I believe that knowledge of Statistics is important for my job.”
    - 1 = Strongly disagree,
    - 2 = Disagree
    - 3 = Neutral
    - 4 = Agree
    - 5 = Strongly agree

Speaker notes

An example of ordinal data is the Likert scale. This takes various forms, but often it is used with group of questions on a questionnaire that reads something like

“Do you agree or disagree with the following statements”

You are asked to respond 1=Strongly disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly agree.

Now I’m sure everyone today is going to choose 5. But assigning numbers 1, 2, 3, 4, and 5 to categories of strongly disagree, disagree, neutral, agree, and strongly agree may falsely imply that a jump from 3 (neutral) to 4 (agree) is about the same amount of improvement as a jump from 4 (agree) to 5 (strongly agree). That’s probably not the case.

You can’t really average ordinal data, some people say because that implies that two responses of “Agree” are the same as one response of “Neutral” along with a response of “Strongly agree”.

Do you want everyone to be at least somewhat on your side or do you want to have a smaller number of very enthusiastic supporters.

If you believe that two 4’s are not the same as a 3 and a 5, then you can’t average.

Now I beg to disagree here, but I am part of a minority opinion. I think that if at the start of this class, your average rating was 3.2 and after I finish the lecture, your average rating climbs to 4.4, that I have done my job well.

If it only jumps to 3.6, then I have still done well, but not as much as that jump to 4.4.

Another example of ordinal data, course grades

- A = 4
- B = 3
- C = 2
- D = 1
- F = 0

Speaker notes

Another example of ordinal data is grades assigned to students. Now everyone in this class is getting an A, but in other classes I teach I might assign different grades. You can attach a number to each of these grades, 4 for A, 3 for B, 2 for C, 1 for D, and 0 for F.

These numbers seem to imply that a student with two B's is as smart as a student with an A and a C.

It raises an interesting story. A colleague of mine told me that he would never hire anyone with a single F on their transcript. An F is a red flag, he felt. So he would not want to assign a value of 0 to F, because that implies that the difference between an F and a D is equivalent to the difference between a B and and A. He's want to assign a value like negative one million to an F so that the average would be pulled way down for a single F, no matter what the other grades would be.

Now I would never be so harsh, but there is really nothing wrong with his perspective. And I would certainly treat a student with three A's and one F differently from a student with two A's and two C's even though mathematically, both average out to 3.0.

Now, in spite of all the obvious problems with equivalence between different grades, most of us still accept a grade point average as a meaningful indicator of how well a student did in school.

Break #4

- What you have learned
  - Scales of measurement
- What's coming next
  - Tests of hypothesis

What is a population?

- Population: a group that you wish to generalize your research results to. It is defined in terms of
  - Demography,
  - Geography,
  - Occupation,
  - Time,
  - Care requirements,
  - Diagnosis,
  - Or some combination of the above.

Speaker notes

A population is a group that you have an interest in. You want to get a better understanding of this group, so you conduct a research study and wish to generalize the results of that study to the population.

In clinical research, a population is almost always a group of people. There are a few exceptions. Sometimes you want to characterize inanimate objects, such as a group of hospitals or a group of medical devices. But let’s keep the focus on people for now.

A population of people is defined in terms of certain characteristics. Usually it is a combination of these characteristics.

Example of a population

All infants born in the state of Missouri during the 1995 calendar year who have one or more visits to the Emergency room during their first year of life.

Speaker notes

Here is an example of a population. It has many of the characteristics described on the previous slide: demography (infants), geography (born in Missouri), time (born in calendar year 1995, during first year of life) and care requirements (one or more ER visits).

Most times the population is so large that it is difficult to get data on all the individuals of that population.

Here, we actually did have access to the data on all 29,637 infants, but most times you would not be so fortunate.

What is a sample?

- Sample: subset of a population.
- Random sample: every person has the same probability of being in the sample.
- Biased sample: Some people have a decreased probability of being in the sample.
  - Always ask “who was left out?”

Speaker notes

A sample is a subset of a population. Because that population of infants was so large, you decided to collect data on a smaller group, a sample of 100 infants, say.

Statistics, according to one definition is the use of data from samples to make inferences about populations. That may be a bit too narrow a definition, but it does characterize quite a bit of what we statisticians do.

A random sample is a special type of sample. It is chosen in a way to insure that every person in the sample has the same probability of being in the sample.

In contrast a biased sample is one where some people in the population have a decreased chance of being in the sample. Often in a biased sample some people in the population are totally excluded.

An example of a biased sample

- A researcher wants to characterize **illicit drug use in teenagers.** She distributes a questionnaire to students attending a local public high school
- (in the U.S. high school is grades 9-12, which is mostly students from ages 14 to 18.)
- Explain how this sample is biased.
- Who has a decreased or even zero probability of being selected.

Type your ideas in the chat box.

Speaker notes

Here is a scenario where a researcher selects a biased sample. I should note here that this is an example specific to the United States. In Italy, you might talk about a survey distributed to the scuola secondaria di secondo grado.

STOP AND GET STUDENT RESPONSES

There are a variety of responses here. The sample does not include home schooled students, students in private schools, students with chronic diseases that force frequent school absences, and students who have dropped out.

Fixing a biased sample

- Redfine your population
  - Not all teenagers,
    - but those attending public high schools.

What is a parameter?

- A parameter is a number computed from a population.
  - Examples
    - Average health care cost associated with the 29,637 children
    - Proportion of these 29,637 children who died in their first year of life.
    - Correlation between gestational age and number of ER visits of these 29,637 children.
  - Designated by Greek letters (μ, π, ρ)

What is a statistic?

- A statistic is a number computed from a sample
  - Examples
    - Average health care cost associated with 100 children.
    - Proportion of these 100 children who died in their first year of life.
    - Correlation between gestational age and number of ER visits of these 100 children.
  - Designated by non-Greek letters ($\bar{X}$, $\hat{p}$, r).

What is Statistics?

- Statistics
  - The use of information from a sample (a statistic) to make inferences about a population (a parameter)
    - Often a comparison of two populations

What is the null hypothesis?

- The null hypothesis (H_0) is a statement about a parameter.
- It implies no difference, no change, or no relationship.
  - Examples
    - $H_0 : \mu_1 - \mu_2 = 0$
    - $H_0 : \pi_1 - \pi_2 = 0$
    - $H_0 : \rho = 0$

What is the alternative hypothesis?

- The alternative hypothesis (H_1 or H_a) implies a difference, change, or relationship.
  - Examples
    - $H_1 : \; \mu_1 - \mu_2 \neq 0$
    - $H_1 : \; \pi_1 - \pi_2 \neq 0$
    - $H_1 : \; \rho \neq 0$

Hypothesis in English instead of Greek

- Only statisticians like Greek letters
  - Translate to simple text
  - For two group comparisons
    - Safer, more effective
  - For regression models
    - Trend, association

Speaker notes

As a researcher, you should always think about your hypothesis in terms of population parameters, but your writing should use text. Translate the Greek letters to English.

If you have a hypothesis that compares two groups, look for comparative words like “safer” or “more effective”. If your hypothesis involves some type of regression model, you should consider terms like “trend” or “association”.

Use PICO

- P = patient population
- I = intervention
- C = control
- O = outcome

Example of text hypotheses, 1 of 2

- "… the objective of this 78-week randomised, placebo-controlled study was to determine whether treatment with nilvadipine sustained-release 8 mg, once a day, was effective and safe in slowing the rate of cognitive decline in patients with mild to moderate Alzheimer disease."
  - Lawlor B, Segurado R, Kennelly S, et al. Nilvadipine in mild to moderate Alzheimer disease: A randomised controlled trial. PLoS Med. 2018; 15(9): e1002660. DOI: 10.1371/journal.pmed.1002660

Speaker notes

Here's an example of a two group comparison. One group gets nilvadipine and the other group gets a placebo. Safety was measured as the proportion of patients who experienced an adverse event. The researchers also measured the proportion of patients who experienced a serious adverse event. So the Greek hypothesis would involve pi's.

Effectiveness was measured using the Alzheimer's Disease Assessment Scale Cognitive Subscale-12 and the Clinical Dementia Rating Scale sum of boxes. Both of these outcome measurements are continuous, so the Greek hypothesis would involve mu's.

PICO for this study

- P = patients with mild to moderate Alzheimer disease
- I = Nilvadine
- C = placebo
- O = cognitive function

Example of text hypotheses, 2 of 2

- "… we investigated trends in BCC incidence over a span of 20 years and the associations between incident BCC and risk factors in a total population of 140,171 participants from 2 large US-based cohort studies: women in the Nurses' Health Study (NHS; 1986–2006) and men in the Health Professionals' Follow-up Study (HPFS; 1988–2006)."
    - Wu S, Han J, Li WQ, Li T, Qureshi AA. Basal-cell carcinoma incidence and associated risk factors in U.S. women and men. Am J Epidemiol. 2013; 178(6): 890–897. DOI: 10.1093/aje/kwt073

Speaker notes

This study used a regression model, a Cox regression model, to study trends and associations, so the Greek hypotheses would involve beta’s.

PICO for this study

- P = female nurses/male health professionals
- I = various risk factors
- C = absence of various risk factors
- O = presence/absence of BCC

One-sided alternatives

- Examples
  - $H_1 : \mu_1 - \mu_2 > 0$
  - $H_1 : \pi_1 - \pi_2 > 0$
  - $H_1 : \rho > 0$
- Changes in only one direction expected
- Changes in opposite direction uninteresting

Passive smoking controversy

- EPA meta-analysis of passive smoking
  - Criticized for using a one-sided hypothesis
  - Samet JM, Burke TA. Turning science into junk: the tobacco industry and passive smoking. Am J Public Health. 2001;91(11):1742–1744.

Speaker notes

Available in html format or PDF format.

Consider a study of the effects of second-hand smoke. These studies always use directional alternatives. From what we know about active cigarette smoking is that it increases the risk of cancer and cardiovascular disease. So there is no reason to expect that passive smoke exposure should be any different than active smoking. Maybe it is less toxic, because of dilution and because the smoking coming off a cigarette from one end is different than the smoke coming off the cigarette from the other end. Fair enough, but there is not reason to believe that things are so different that all of a sudden the smoke becomes protective.

Since there is no scientific basis for a protective effect of passive smoking, it makes sense to test that passive smoking has no effect versus it having an increase in bad outcomes compared to the control group. So your null hypothesis is “not harmful” and your alternative is “harmful”. The beneficial hypothesis is lumped into the null hypothesis, but no one would dare claim that passive smoking was protective.

Actually, the tobacco companies did complain that the use of a directional alternative violated the norms of science. They won in a court battle in North Carolina, but lost on appeal.

As another aside, I was involved with prayer study. We planned this study using a one-sided hypothesis (remote prayer has a positive effect on health). The Institutional Review Board suggested changing this to a two-sided hypothesis (remote prayer has either a positive or a negative effect on health). Thankfully, we did not observe an outcome in the opposite tail as that would have been very difficult to explain.

What is a decision rule? 1 of 3

- Example
  - $H_0\ :\ \mu_1 - \mu_2 = 0$
  - $H_1\ :\ \mu_1 - \mu_2 \neq 0$
  - t = $(\bar{X}_1 - \bar{X}_2)$ / se
  - Accept H_0 if t is close to zero.

What is a decision rule? 2 of 3

- Example
  - $H_0 : \pi_1 - \pi_2 = 0$
  - $H_1 : \pi_1 - \pi_2 \neq 0$
  - t = $(\hat{p}_1 - \hat{p}_2)$ / se
  - Accept H_0 if t is close to zero.

What is a decision rule? 3 of 3

- Example
  - $H_0 : \ \rho = 0$
  - $H_1 : \ \rho \neq 0$
  - t = r / se
  - Accept H_0 if t is close to zero.

What is a Type I error?

- A Type I error is rejecting the null hypothesis when the null hypothesis is true
  - False positive
  - Example involving drug approval: a Type I error is allowing an ineffective drug onto the market.
- α = P[Type I error]

Speaker notes

In your research, you specify a null hypothesis (typically labeled H0) and an alternative hypothesis (typically labeled Ha, or sometimes H1). By tradition, the null hypothesis corresponds to no change. When you are using Statistics to decide between these two hypothesis, you have to allow for the possibility of error. Actually, if you are using any other procedure, you should still allow for the possibility of error, but we statisticians are the only ones honest enough to admit this.

A Type I error is rejecting the null hypothesis when the null hypothesis is true.

Consider a new drug that we will put on the market if we can show that it is better than a placebo. In this context, H0 would represent the hypothesis that the average improvement (or perhaps the probability of improvement) among all patients taking the new drug is equal to the average improvement (probability of improvement) among all patients taking the placebo. A Type I error would be allowing an ineffective drug onto the market.

Remember that the hypotheses involve population parameters. Population parameters are impossible to compute. So you can only talk about Type I errors in an abstract sense. You will never know for certain if you have made a Type I error.

Alpha is the probability of a Type I error, and alpha is a value that you can compute. In most studies, researchers work hard to keep the probability of a Type I error low, typically at 5%.

What is a Type II error?

- A Type II error is accepting the null hypothesis when the null hypothesis is false.
  - False negative result
  - Usually computed at MCD
  - An example involving drug approval: a Type II error is keeping an effective drug off of the market.
- β = P[Type II error]
- Power = $1 - \beta$

Speaker notes

A Type II error is accepting the null hypothesis when the null hypothesis is false. You should always remember that it is impossible to prove a negative. Some statisticians will emphasize this fact by using the phrase “fail to reject the null hypothesis” in place of “accept the null hypothesis.” The former phrase always strikes me as semantic overkill.

Many studies have small sample sizes that make it difficult to reject the null hypothesis, even when there is a big change in the data. In these situations, a Type II error might be a possible explanation for the negative study results.

Consider a new drug that we will put on the market if we can show that it is better than a placebo. In this context, H0 would represent the hypothesis that the average improvement (or perhaps the probability of improvement) among all patients taking the new drug is equal to the average improvement (probability of improvement) among all patients taking the placebo. A Type II error would be keeping an effective drug off the market.

It bears repeating that population parameters are impossible to compute. So you will never know for certain if you have made a Type I error.

Beta is the probability of a Type II error. Beta is a known quantity. Typically researchers try to keep beta small. 10% is a typical value, though in some settings, a Type II error rate as large as 20% could be tolerated.

Power is defined as 1-beta. I will talk more about power in a little bit.

What is a p-value?

- Let t =
  - $(\bar{X}_1 - \bar{X}_2)$ / se, or
  - $(\hat{p}_1 - \hat{p}_2)$ / se, or
  - r / se
- p-value = Prob of sample result, t, or a result more extreme,
  - **assuming the null hypothesis is true**
- Small p-value, reject H_0
- Large p-value, accept H_0

Speaker notes

A p-value is a measure of how much evidence we have against the null hypothesis.

The smaller the p-value, the more evidence we have against H0.

The p-value is also a measure of how likely we are to get a certain sample result or a result “more extreme,” assuming H0 is true.

The type of hypothesis (right tailed, left tailed or two tailed) will determine what “more extreme” means.

Alternate interpretations

- Consistency between the data and the null
  - Small value, inconsistent
  - Large value, consistent
- Evidence against the null
  - Small, lots of evidence against the null
  - Large, little evidence against the null

Speaker notes

There are two interpretations that I feel are more practical. You can think of the p-value as a measure of consistency between the data and the null hypothesis. A small value implies inconsistency. It is very unlikely that you will get a value like you've seen in your sample or a value more extreme under the assumption that the null hypothesis is true. So you should reject that assumption.

On the other hand if the sample results or anything more extreme has a high probability under the assumption that the null hypothesis is true, then you should feel comfortable accepting that assumption.

I have argued that the p-value is a measure of evidence. Some have called it a poor measure of evidence, but I stand by my interpretation.

If the p-value is small, you have lots of evidence against the null hypothesis. If the p-value is large, you have little or no evidence against the null hypothesis.

What the p-value is not, 1 of 2

- A p-value is NOT the probability that the null hypothesis is true.
  - P[t or more extreme | null] is different than
  - P[null | t or more extreme]
    - P[null] is nonsensical
    - μ, π, or ρ are unknown constants (no sampling error)

Speaker notes

The p-value is a conditional probability, and you always need to be careful about conditional probabilities. It is a probability about a sample result given an assumption about the population result. It is not a probability about a population result given the sample result. There are two reasons for this.

First, you can't reorder a conditional probability. The probability of A given B is almost never the same as the probability of B given A. The example I give for this is the probability of being happy given that you are rich. That's a pretty high number, I hope you'll agree. There are a few rich people who lead miserable lives, but from everything I've seen, most rich people are pretty darn happy. The reverse of this is the probability of being rich given that you are happy. That number is much smaller. Because although I believe that money can buy happiness, a lot of other things can also buy happiness just as well. It's not quite as easy to find happiness if you're poor, but somehow, a lot of poor people find a way to be happy anyway.

A second reason that you can't reverse the order is that you cannot make a probability statement about population parameters. They are numbers computed from the entire population, and are fixed values. You cannot make a probability statement about something that has no sampling error.

Only numbers computed from a sample (i.e., statistics) have sampling error.

What the p-value is not, 2 of 2

- Not a measure FOR either hypothesis
  - Little evidence **against** the null $\neq$ lots of evidence **for** the null
- Not very informative if it is large
  - Need a power calculation, or
  - Narrow confidence interval
- Not very helpful for huge data sets

Speaker notes

The p-value is not a measure for either hypothesis. It is always a measure against a particular hypothesis. Now when the p-value is small, you can make a strong statement. We have lots of evidence against the null hypothesis. That translates into lots of evidence in favor of the alternative hypothesis.

When the p-value is large, however, you are in a quandary. Little or no evidence against the null hypothesis is not the same as lots of evidence for the null hypothesis.

It's possible to have little or no evidence against the null and also have little or no evidence against the alternative. This happens whenever you have a really small sample size combined with a lot of noise.

You can't prove a negative, so the saying goes. Well, you can prove a negative, but you have to work harder at it. A large p-value by itself is not persuasive, but if you combine it with a power calculation done prior to data collection, that's pretty good evidence in support of the null hypothesis.

You could also combine a large p-value with a narrow confidence interval to support the null hypothesis. I'll talk about that more in just a bit.

In general, the p-value is not very helpful for large samples. We're seeing this more and more. Just about everything pops up as statistically significant with these huge data sets, and you can't use the p-value to separate the important stuff from the trivial stuff. You need to look instead at the magnitude of the sample estimates and calculate how much uncertainty you can remove in your future predictions.

A bad test question

A research paper computes a p-value of 0.45. How would you interpret this p-value?

1. Strong evidence **for** the null
2. Strong evidence **for** the alternative
3. Little or no evidence **for** the null
4. Little or no evidence **for** the alternative
5. More than one answer above is correct.
6. I do not know the answer.

Speaker notes

Here’s that pop quiz again. Take a look at it quickly. Note that the p-value is of evidence against the null hypothesis. So each of the first four responses is wrong.

I wrote this question quickly, so shame, shame on me. But I’ve reproduced the example because it illustrates an important point.

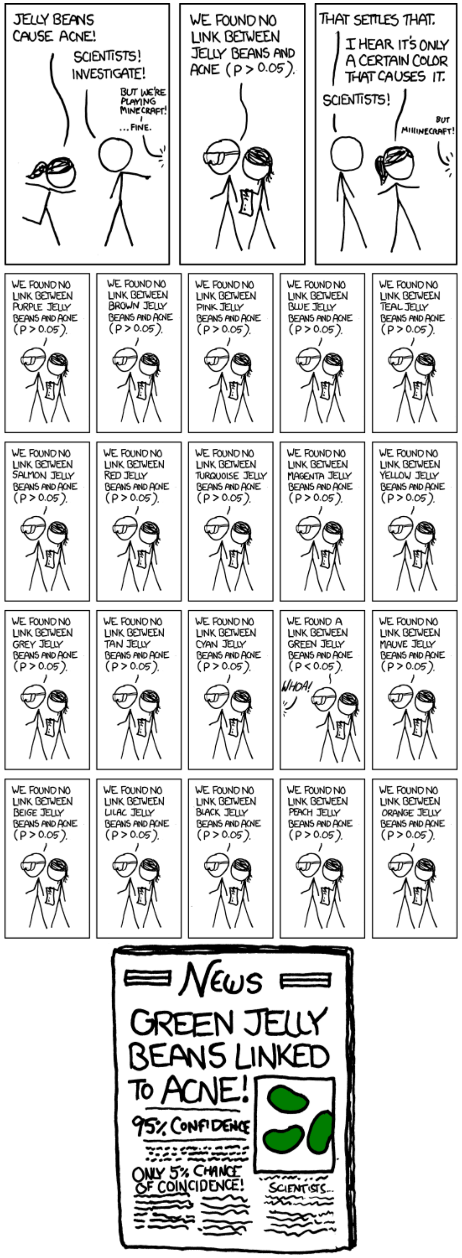


Figure 1: xkcd cartoon about jelly beans and cancer

Speaker notes

This cartoon is impossible to read, but you can find it on the Canvas site or in the readings. Here’s a brief run down.

In the first panel, a woman runs up to a man and shouts: Jelly beans cause acne!

The man replies : Scientists! Investigate!

In the second panel, one scientist, holding a clipboard announces: We found no link between jelly beans and acne ($p > 0.05$).

In the third panel, the woman says: I hear it’s only a certain color that causes it.

In a bunch of small panels, the scientist with a clipboard reports: We found no link between purple jelly beans and acne ($p > 0.05$).

We found no link between brown jelly beans and acne ($p > 0.05$).

We found no link between pink jelly beans and acne ($p > 0.05$).

The same for blue, teal, salmon, red, and so forth. And then…

We found a link between green jelly beans and acne ($p < 0.05$). An off-screen voice goes: Whoa!

The next six panels show

We found no link between mauve jelly beans and acne ($p > 0.05$).

We found no link between beige jelly beans and acne ($p > 0.05$).

We found no link between lilac jelly beans and acne ($p > 0.05$).

We found no link between black jelly beans and acne ($p > 0.05$).

We found no link between peach jelly beans and acne ($p > 0.05$).

We found no link between orange jelly beans and acne ($p > 0.05$).

At the bottom is a newspaper with the headline: Green Jelly Beans Linked To Acne! 95% Confidence. Only 5% chance of coincidence!

If you are interested in a transcript and a detailed explanation, https://www.explainxkcd.com/wiki/index.php/882:_Significant

What is p-hacking?

- Abuse of the hypothesis testing framework.
  - Run multiple tests on the same outcome
  - Test multiple outcome measures
  - Remove outliers and retest
- Defenses against p-hacking
  - Bonferroni
  - Primary versus secondary
  - Published protocol

Speaker notes

This is an example of p-hacking. You change the testing process to increase the probability of a Type I error (Rejecting the null hypothesis when the null hypothesis is true). This increases the chance of getting a positive result, which you may find desirable, but only by increasing the probability of a false positive result.

Some examples of p-hacking. Run multiple tests on the same outcome measure. Start with the regular t-test, include the t-test that allows for unequal variances, and run two different non-parametric tests, the Wilcoxon-Mann-Whitney test and the sign test. Choose the test with the smallest p-value.

You also might consider multiple outcome measures. Compare the mortality rate, the relapse rate, and the re-hospitalization rate. If any of the three is statistically significant, claim victory.

You could also do this with longitudinal data. Compare pain relief at one hour and at four hours. If you see a difference at one hour, claim that your new medication is faster acting. If you see a difference at four hours, claim that your medication is longer lasting.

You might run a test with the full data set and then with an outlier or two removed. Report for the data set that has the smaller p-value and pretend that this was your original choice all along.

These are only a few of the choices. I don’t want to say more because I feel like I’m the devil tempting you.

There are two defenses against p-hacking. Well three if you count being honest. But what I mean is there are two things that you can do that will satisfy others that you are playing fairly.

First, you can adjust your decision rule by using a Bonferroni correction. Bonferroni divides alpha by the number of tests. If you are using three different outcome measures, compare your p-value of 0.0133 instead of 0.05.

Second, you can designate one of your outcome measures as primary. If you achieve statistical significance on your primary outcome, great. The remaining outcome measures are secondary. If you achieve statistical significance on a secondary outcome meausure only, report the results as provisional and requiring independent replication.

You should publish a detailed protocol, either through a clinical trial registry, or now there are journals which accept publications of the research protocols before any data are collected. It’s a paper with literature review and methods section, but no results and no discussion section.

Now p-hacking has happened because some people have a skewed view of research. They are interested in using research to promote their own agenda rather than using research to uncover the truth. Perfectly understandable if you are a drug company, but you as an independent researcher should never try to skew the data. It hurts you and it hurts your patients. You need to adopt a disinterested posture in that you are glad when the research points in one direction and you are glad when it points in the opposite direction, because either way, you know more than you did before and you can treat your patients better because of this knowledge.

Break #5

- What you have learned
  - Tests of hypothesis
- What's coming next
  - Confidence intervals

What is a confidence interval?

- Range of plausible values
  - Tries to quantify uncertainty associated with the sampling process.

Speaker notes

We statisticians have a habit of hedging our bets. We always insert qualifiers into our reports, warn about all sorts of assumptions, and never admit to anything more extreme than probable. There's a famous saying: "Statistics means never having to say you're certain."

We qualify our statements, of course, because we are always dealing with imperfect information. In particular, we are often asked to make statements about a population (a large group of subjects) using information from a sample (a small, but carefully selected subset of this population). No matter how carefully this sample is selected to be a fair and unbiased representation of the population, relying on information from a sample will always lead to some level of uncertainty.

A confidence interval is a range of values that tries to quantify uncertainty associated with the sampling process.

Consider it as a range of plausible values.

There is a confidence level associated with any confidence interval, usually 95%, but sometimes 90% or 99%.

The confidence level is related to the alpha level (probability of a Type I error).

It also has a long range sampling interpretation.

If you repeatedly sampled from the same population, then 95% (or 90% or 99%) of the confidence intervals produced would contain the true value in the population.

Example of a confidence interval

- Homeopathic treatment of swelling after oral surgery
  - 95% CI: -5.5 to 7.5 mm
  - Lokken P, Straumsheim PA, Tveiten D, Skjelbred P, Borchgrevink CF. Effect of homoeopathy on pain and other events after acute trauma: placebo controlled trial with bilateral oral surgery BMJ. 1995;310(6992):1439-1442.

Speaker notes

You can find this paper at

http://www.bmj.com/content/310/6992/1439.full

Always look for confidence intervals that are wide enough to drive a truck through. They are very good indicators of small sample sizes.

Consider a recent study of homoeopathic treatment of pain and swelling after oral surgery (Lokken 1995). When examining swelling 3 days after the operation, they showed that homoeopathy led to 1 mm less swelling on average. The 95% confidence interval, however, ranged from -5.5 to 7.5 mm. From what little I know about oral surgery, this appears to be a very wide interval. This interval implies that neither a large improvement due to homoeopathy nor a large decrement could be ruled out.

Now, you can’t drive a truck through an interval that goes from -5.5 to 7.5 mm, but from the perspective of a human mouth, this interval is huge. Generally when a confidence interval is very wide like this one, it is an indication of an inadequate sample size, an issue that the authors mention in the discussion section of this paper.

Confidence interval interpretation (1 of 7)

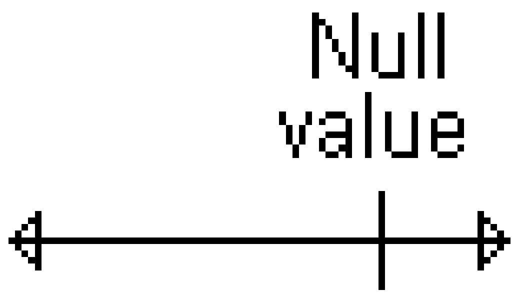


Figure 2: Interval that contains the null value

Speaker notes

When you see a confidence interval in a published medical report, you should look for two things. First, does the interval contain a value that implies no change or no effect? For example, with a confidence interval for a difference look to see whether that interval includes zero. With a confidence interval for a ratio, look to see whether that interval contains one.

Here’s an example of a confidence interval that contains the null value. This interval implies no statistically significant change.

Confidence interval interpretation (2 of 7)

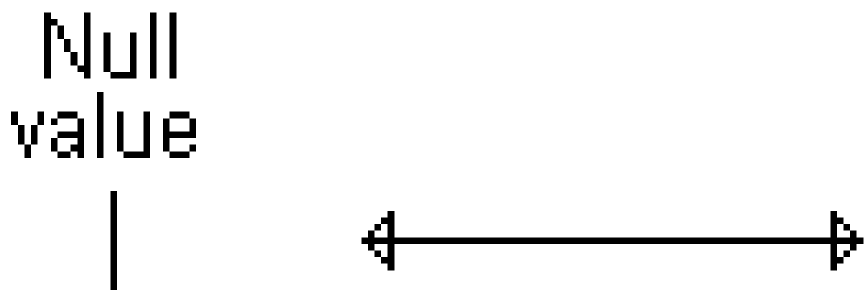


Figure 3: Interval entirely above the null value

Speaker notes

Here’s an example of a confidence interval that excludes the null value. If we assume that larger implies better, then the interval would imply a statistically significant improvement.

Confidence interval interpretation (3 of 7)

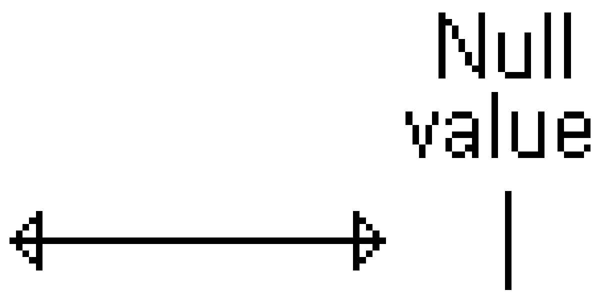


Figure 4: Interval entirely below the null value

Speaker notes

Here’s a different example of a confidence interval that excludes the null value. This interval implies a statistically significant decline.

Confidence interval interpretation (4 of 7)

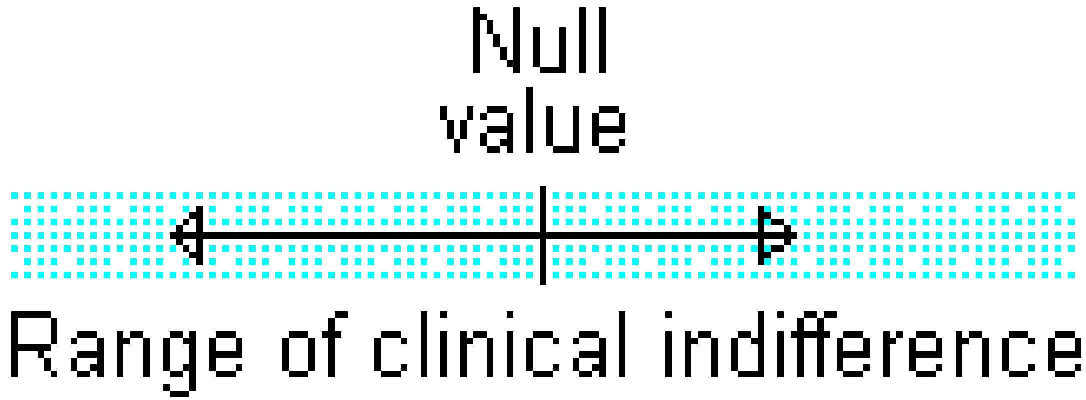


Figure 5: Interval entirely inside the range of clinical indifference

Speaker notes

You should also see whether the confidence interval lies partly or entirely within a range of clinical indifference. Clinical indifference represents values of such a trivial size that you would not want to change your current practice. For example, you would not recommend a special diet that showed a one year weight loss of only five pounds. You would not order a diagnostic test that had a predictive value of less than 50%.

Clinical indifference is a medical judgment, and not a statistical judgment. It depends on your knowledge of the range of possible treatments, their costs, and their side effects. As statistician, I can only speculate on what a range of clinical indifference is. I do want to emphasize, however, that if a confidence interval is contained entirely within your range of clinical indifference, then you have clear and convincing evidence to keep doing things the same way.

Confidence interval interpretation (5 of 7)

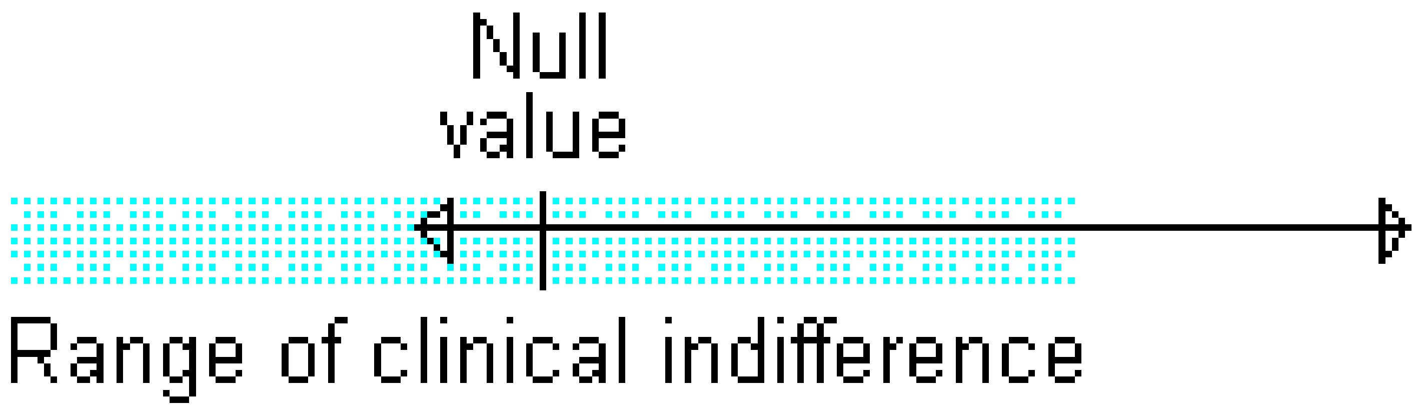


Figure 6: Interval partly inside/outside range of clinical indifference

Speaker notes

One the other hand, if part of the confidence interval lies outside the range of clinical indifference, then you should consider the possibility that the sample size is too small.

The interval contains zero, so it is plausible to behave as if the difference in population means or proportions is zero. But the interval also contains values that are clinically important. So it is plausible to behave as if there is a clinically important difference in means. How can you have two such different interpretations being plausible at the same time? That's the definition of ambiguity. If you don't like it, get used to it. Statistics will often identify areas of ambiguity, which is a good thing, because it tells us to not act prematurely, but instead demand more data before you make a definitive decision.

Quiz question, revisited

A research paper computes a confidence interval for a relative risk of 0.82 to 3.94. This confidence interval tells that the result is

1. statistically significant and clinically important.
2. not statistically significant, but is clinically important.
3. statistically significant, but not clinically important.
4. not statistically significant, and not clinically important.
5. The result is ambiguous.
6. I do not know the answer.

Speaker notes

Let’s go back to that question I posed earlier.

A research paper computes a confidence interval for a relative risk of 0.82 to 3.94. What does this confidence interval tell you.

Well, it tells you that a relative risk of 1 (equal risks) is plausible, but that a relative risk of 2 (a doubling of risk) is also plausible. A tripling of risk is plausible. Good grief! This is an ambiguous result.

Doesn’t this bother you? It should. Someone ran a terrible experiment. An experiment so poorly designed that it can’t distinguish between no change in risk, or a tripling of risk.

It’s a terrible thing, but it happens all the time and it doesn’t seem to bother anyone but me. This is wretched. You got a hundred patients to let you poke and prod them. They took some bitter pills or maybe placebos. They are sacrificing their bodies in the name of science. And the best you can do is a confidence interval that goes from 0.82 to 3.94. Hang your head in shame!

Confidence interval interpretation (6 of 7)

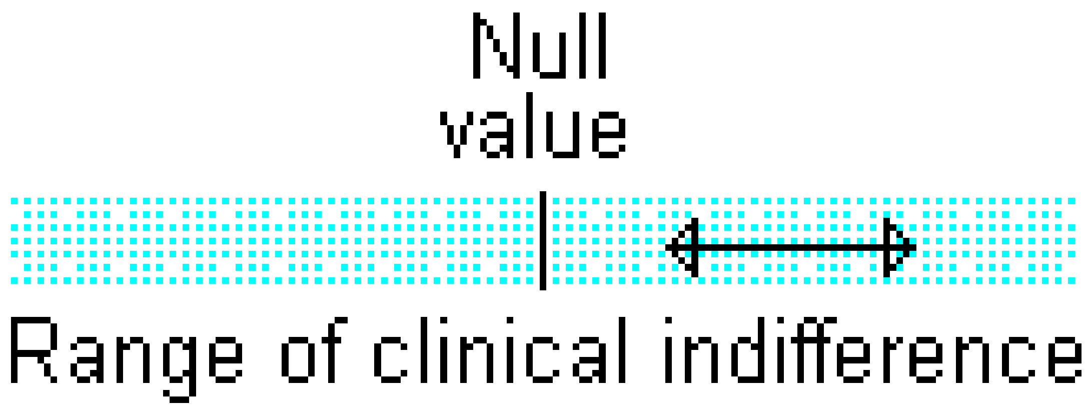


Figure 7: Confidence interval entirely inside the range of clinical indifference

Speaker notes

Some studies have sample sizes that are so large that even trivial differences are declared statistically significant, especially in this era of big data. If your confidence interval excludes the null value but still lies entirely within the range of clinical indifference, then you have a result with statistical significance, but no practical significance.

Confidence interval interpretation (7 of 7)

Figure 8: Confidence interval entirely outside the range of clinical indifference

Speaker notes

Finally, if your confidence interval excludes the null value and lies outside the range of clinical indifference, then you have both statistical and practical significance.

Let's talk about one more case. I don't have a picture, but imagine a confidence interval that is mostly in the white region, the region of clinical importance, but the lower limit stretches into the range of clinical indifference. It doesn't quite include the null value, but it comes within kissing distance. That's a result that achieves statistical significance, but it does not provide definitive proof of clinical importance. No one ever talks about this case, but they should. Your confidence interval indicates statistical significance, but just barely. So don't pretend that your results are the final word. You should not stop researching until you get a confidence interval that lies entirely inside or entirely outside the range of clinical indifference.

Why you might prefer a confidence interval

- Provides same information as p-value,
  - Clinical importance
  - Distinguish between
    - definitive negative result, or
    - more research is needed

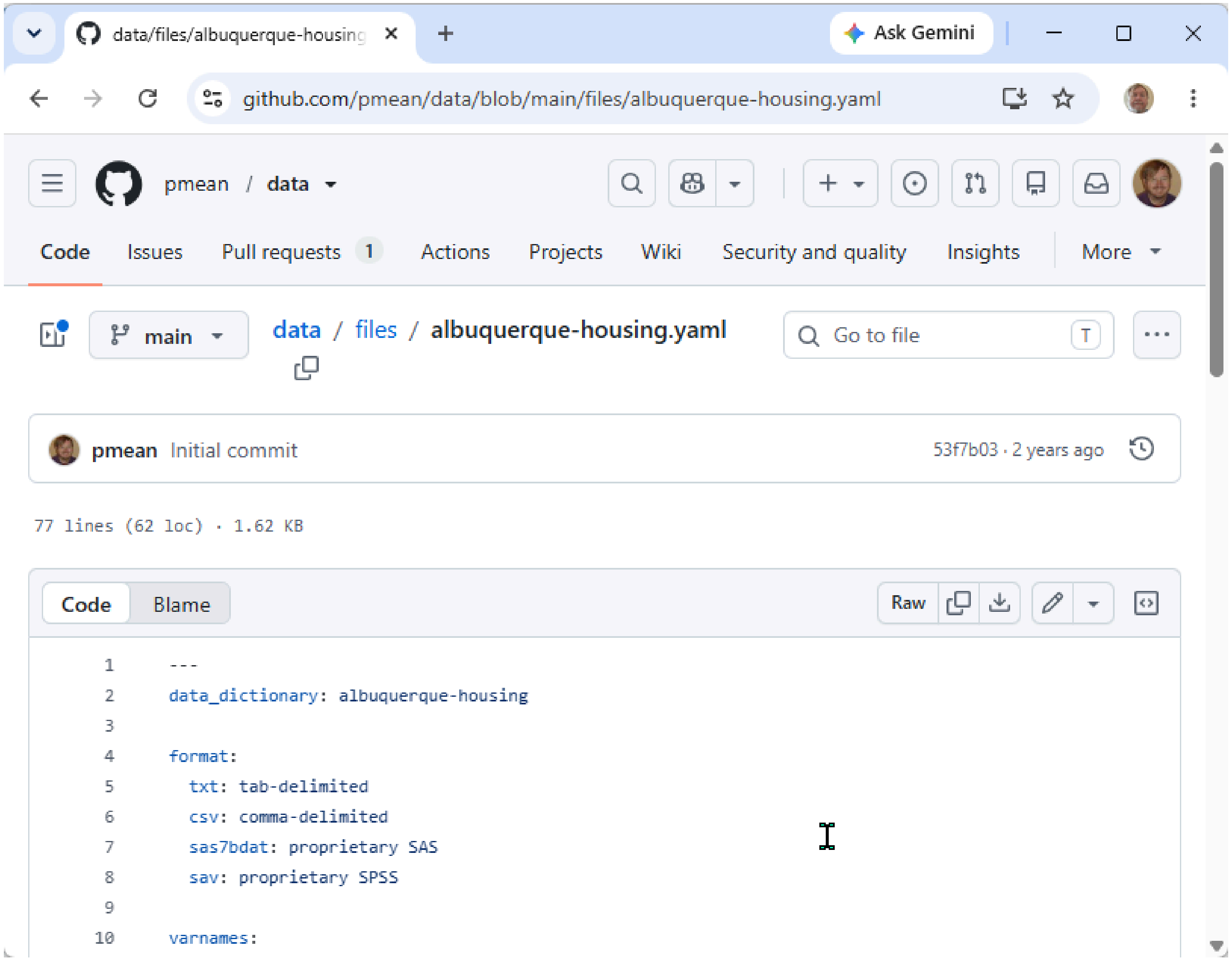
Speaker notes

Add notes here.

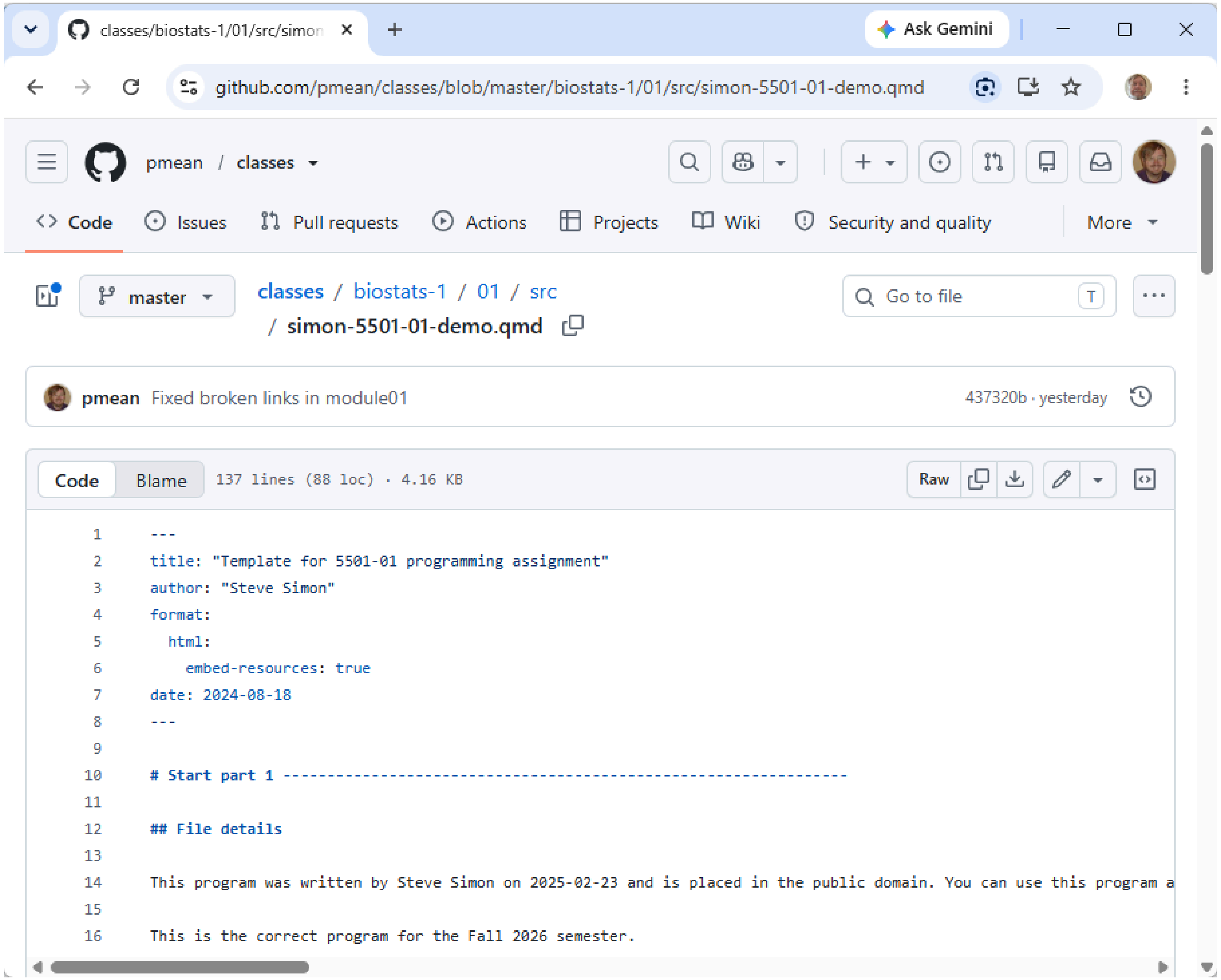
Break #6

- What you have learned
  - Confidence intervals
- What's coming next
  - A simple R program

albuquerque-housing.yaml



simon-5501-01-demo.qmd

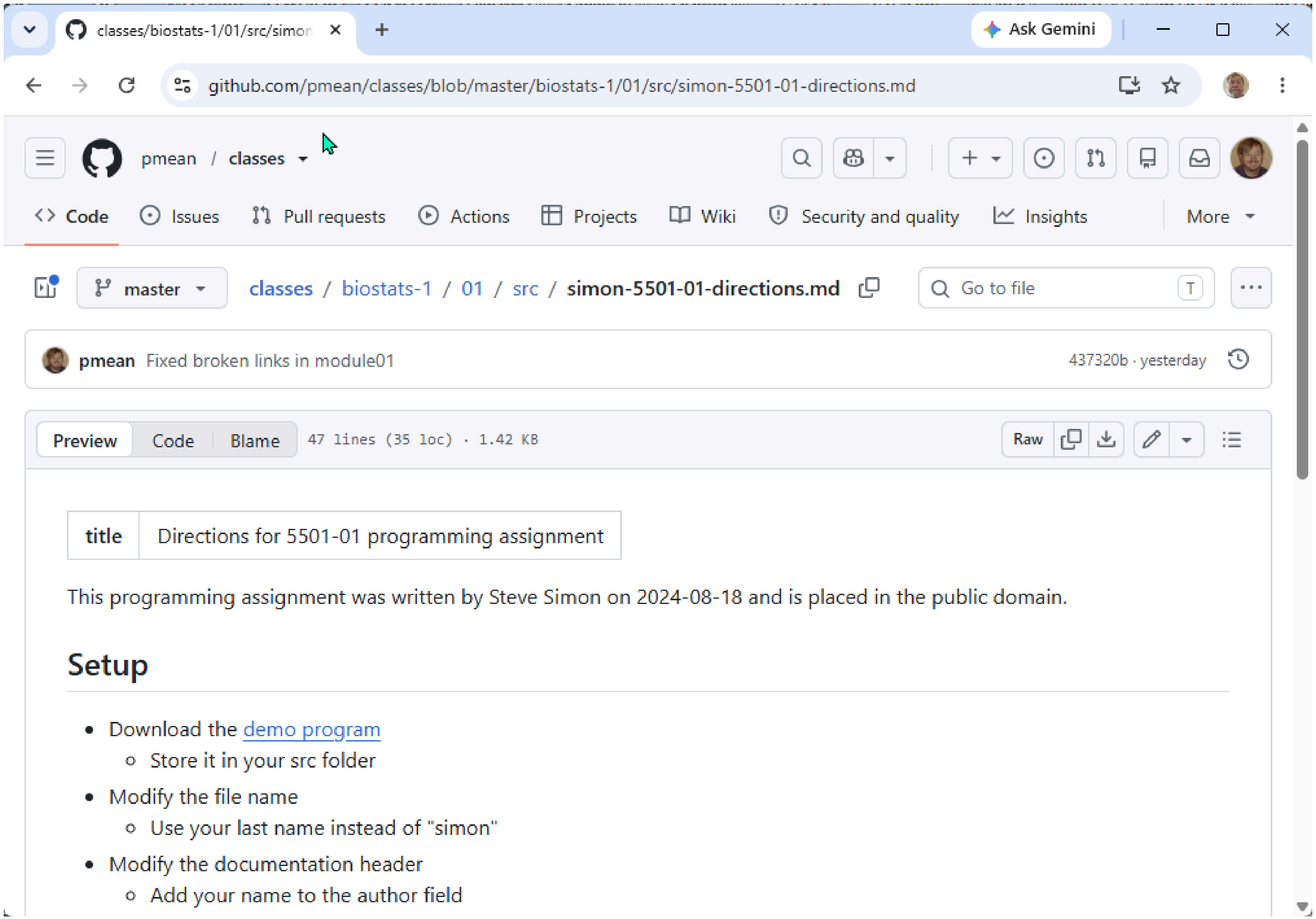


Live demo

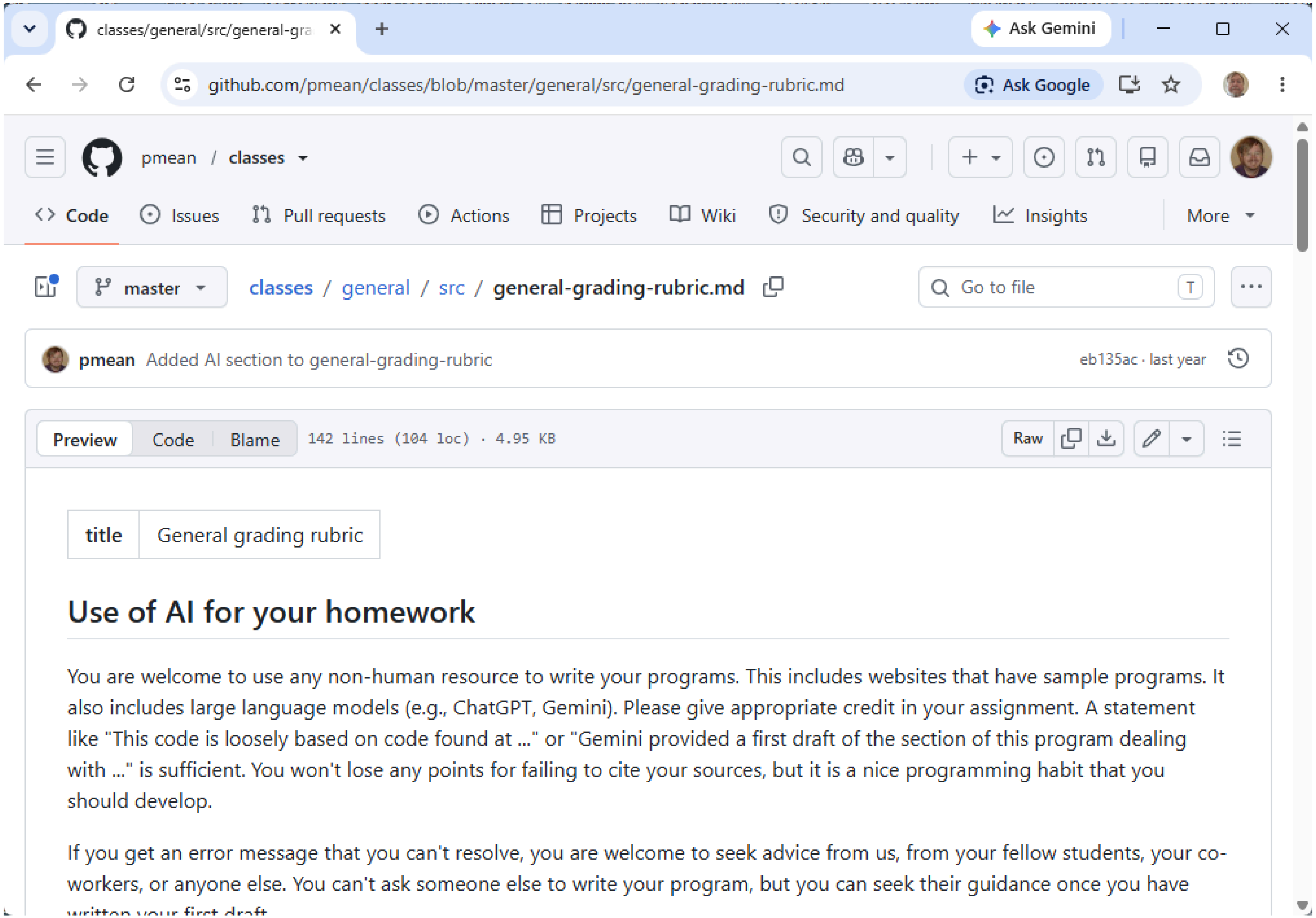
Break #7

- What you have learned
  - A simple R program
- What's coming next
  - Your homework

simon-5501-01-directions



general-grading-rubric



Live demo

Summary

- What you have learned
    - About this class
    - R and RStudio
    - History of R
    - Scales of measurement
    - Tests of hypothesis
    - Confidence intervals
    - A simple R program
    - Your homework